

Smart Campus IoT Simulation and Analytics Platform

Project Proposal

Course: Internet of Things | Program: Master of IT
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1. Problem

Modern campuses contain classrooms, labs, offices, and server rooms that need continuous monitoring for comfort, safety, and energy efficiency. Building a full hardware IoT prototype is costly and time-consuming for a course project. This project solves that by creating a simulation-only IoT platform that demonstrates the complete IoT workflow without physical sensors.

2. Scenario

Smart Campus Environment Monitoring with 7 virtual devices across a university campus:

Device ID	Label	Building	Purpose
hall_1_1	Hall 1.1	floor_1	Teaching environment monitoring
hall_1_2	Hall 1.2	floor_1	Teaching environment monitoring
hall_1_3	Hall 1.3	floor_1	Teaching environment monitoring
hall_1_4	Hall 1.4	floor_1	Teaching environment monitoring
server_room	Server Room	it_building	High-temperature risk monitoring
lab1	Lab 1	engineering_building	Lab comfort & air quality
office1	Office 1	admin_building	Office energy & occupancy

3. Selected Tools

Tool	Purpose
Python 3	Virtual device simulator and MQTT subscriber
Paho MQTT	Python MQTT client library
HiveMQ Broker	Public MQTT broker for testing
Node-RED	Dashboard for live visualization
python-docx	Report generation

4. Wireless Communication Model

Selected model: IEEE 802.15.4 / Zigbee-like.

Justification: Campus room sensors send small packets at low frequency and may be battery-powered. Zigbee-like communication is ideal for short-range indoor sensor networks with low power requirements.

The simulator includes: packet loss rate, random delay, RSSI, link quality, estimated energy consumption, and battery drain per transmission.

5. System Architecture

1. Virtual Device Layer → Python devices generate simulated sensor readings.
2. Communication Layer → MQTT topics transfer telemetry from devices to subscribers.
3. Processing Layer → Python subscriber validates, computes derived values, and triggers alerts.
4. Presentation Layer → Node-RED Dashboard displays live values and trends.

6. MQTT Topic Design

```
campus/{building}/{room}/{deviceId}/telemetry
```

Example:

```
campus/floor_1/hall_1_1/hall_1_1/telemetry
```

7. Expected Deliverables

- Runnable Python simulator (7 virtual devices).
- MQTT subscriber with validation, analytics, and alerts.
- Node-RED Dashboard flow JSON.
- Test cases documentation.
- Final technical report (8–12 pages).
- Demo video (5–8 minutes).
- Short presentation (5–7 slides).